INCH-POUND

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DETAIL SPECIFICATION

CASE, MAINTENANCE EQUIPMENT, SMALL ARMS, M16A1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the design, materials and fabrication of a case for carrying maintenance equipment for the M16A1 Small Arms.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support, Standardization Office, 700 Robbins Avenue, Philadelphia, PA 19111-5092 or emailed to trpsptctstandardiz@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8465

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-55126	-	Fastener Tapes, Hook and Loop, Synthetic
A-A-59963	-	Thread, Polyester

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-4088	-	Webbing, Textile, Woven Nylon
MIL-DTL-5038	-	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-C-7219	-	Cloth, Duck, Nylon
MIL-H-9890	-	Hardware, Individual Load Carrying Equipment: and Hardware, Miscellaneous
MIL-DTL-10884	-	Fasteners, Snap
MIL-DTL-27265	-	Webbing, Textile, Woven Nylon Impregnated
MIL-DTL-32075	-	Label: for Clothing, Equipage, and Tentage, (General Use)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

(Copies of the documents are available online at https://quicksearch.dla.mil/.)

DRAWINGS

NATICK LABORATORY ARMY RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-2-282	-	Case, Maintenance Equipment, Small Arms, (For M16A1 Rifle); Assembly Complete

(Copies of drawings are available from the U.S. Army, DEVCOM Soldier Center, ATTN: FCDD-SCP-PE, 10 General Greene Avenue, Natick, MA 01760-5019.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AEROSPACE INDUSTRY ASSOCIATION OF AMERICA INC.

NASM 27981 - Fasteners, Snap, Style 2A (Small Wire Spring Clamp Type)

(Copies of this document are available from https://www.aia-aerospace.org)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of this document are available online at https://asq.org)

ASTM International

ASTM D6193 - Standard Practice for Stitches and Seams

(Copies of these documents are available online at https://www.astm.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4). Guide samples of the end item, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from the specification may appear in the sample, in which case the specification shall govern.

3.2 Standard sample. Unless otherwise indicated, the finished CASE shall match the standard sample for shade and appearance, and shall, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials.

3.4.1 Cloth, duck, nylon. The nylon cloth shall be Camouflage Green 483 and shall conform to Type III, Class 3 of MIL-C-7219.

3.4.2 Webbing, nylon. The nylon webbing shall be Camouflage Green 483 and shall conform to Type VIIIc, Class 2 of MIL-DTL-4088 and class R of MIL-DTL-27265.

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3.4.3 Tape, nylon. The nylon tape used for binding shall be dyed Camouflage Green 483 and shall conform to Type III, ¾ inch wide of MIL-DTL-5038.

3.4.4. Fastener tape, hook and pile, nylon. The nylon fastener tape shall be dyed Camouflage Green 483 and shall conform to Type II, Class 1, ⅝ inch wide of A-A-55126.

3.4.5 Thread, polyester. The thread for all stitching shall conform to Type I, Class 1, sub-class A or B, size E of A-A-59963.

3.4.5.1 Color. The thread shall be dyed Camouflage Green 483 and shall show fastness to weathering equal to or better than the standard sample.

3.4.6 Keeper, with slide. The keeper with slide shall conform to Type X of MIL-H-9890.

3.4.7 Fasteners, snap. The snap fastener shall be style 2A, finish 2 conforming to MIL-DTL-10884 and the following listed part numbers of NASM 27981:

Component	Part Number
Button	-1B
Socket	-3B
Stud	-4B
Eyelet	-5B

3.5 Construction. The construction shall conform in all respects to Drawing 2-2-282 and as specified herein.

3.5.1 Stitching, machine. All stitching, except bartacking, shall conform to type 301 of ASTM D6193 with 8 to 10 stitches per inch.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or over-stitched ½ inch minimum except where ends are turned under in a hem or held down by other stitching. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.5.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. Unless otherwise specified, when thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of ½ inch back of the end of the stitching. 1/

b. Unless otherwise specified, thread breaks, or two or more consecutive skipped or runoff stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of ½ inch in back of the defective area, continue over the defective area and continue a minimum of ½ inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by

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removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Bartacking. Unless otherwise specified, bartacking shall be ½ (± 1/16) inch in length, ⅛ (± 1/32) inch in width and shall contain 28 stitches. Bartacking shall be free from thread breaks and loose stitching.

3.5.1.3 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

3.5.1.4 Thread ends. All thread ends shall be trimmed to a length of not more than ¼ inch.

3.5.2 Setting of snap fasteners. A hole shall be prepunched through the materials before insertion of the button barrel of the female component and the eyelet barrel of the male component. The hole shall be smaller than the outside diameter of the button barrel or the eyelet barrel, as applicable, so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fastener components shall be securely clinched without cutting the adjacent materials and no more than three splits shall occur in the button or eyelet barrels.

3.5.3 Fusing of nylon tape and webbing ends. All ends of nylon tape and webbing shall be fused. The apparatus used to fuse the tape and webbing ends shall be capable of providing sufficient heat to provide a smooth edge with the cut ends of the tape and webbing yarns all fused together.

3.5.4. Location marks. Location marks shall not be drilled except for locating snap fasteners.

3.5.5 Repairs. Repairs such as mends, darns or patches shall not be made to the case.

3.5.6 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3.1 shall be removed from production and replaced with non-defective and properly matched components.

3.6 Marking. The identification marking shall be applied in the location shown on the drawing and shall conform to Type IV, Class 5 of MIL-DTL-32075. The nomenclature may be abbreviated to read "Case, Maint. Equip., M16A1 Rifle". The letters "US" shall be applied in the location and in the size characters indicated on Drawing 2-2-282 and shall conform to Type IV, Class 9 of MIL-DTL-32075. Fastness of the Class 9 marking shall be as specified for Class 5 marking.

3.7 Workmanship. The finished case shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3.1 and 4.4.3.2. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether construction details which cannot be examined on the finished item are in accordance with specified requirements. This inspection shall include verification that holes punched for snap fasteners conform to 3.4.2. Whenever nonconformance is noted, corrections shall be made to the items affected and lot in-process. Items which cannot be corrected shall be removed from production.

4.4.3 End item examination. The end item shall be examined in accordance with 4.4.3.1, 4.4.3.2 and 4.4.3.3. The lot size shall be expressed in units of one (1) case. The sample unit shall be one (1) case.

4.4.3.1 Visual examination. The end item shall be examined for the defects listed below in Table I. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits as specified in the contract and/or procurement documents, except where otherwise indicated.(see 6.5).

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Table I: Classification Of Defects

Examine	Defect	Classification	
		Major	Minor
Cloth	Any hole, cut, tear, or abrasion.	101	
	Smash, open place, broken or missing yarn, or multiple floats.	102	
Webbing and tape	Cut ends not fused or not fused correctly, i.e., yarns not fused together or not smooth.		201
	Frayed or scalloped edges; not firmly and tightly woven		202
Fastener tape, hook and pile	Any hole, cut, or tear	103	
	Hooks missing or flattened impairing function.	104	
Hardware: General			
	Broken or malformed, corroded area, burr, or sharp edge.	105	
	Finish omitted.	106	
	Finish defects, i.e., wrinkles, drops, thin or loose film.		203
Keeper	Slide component of keeper jams in open, partly closed, or closed position.	107	
	NOTE: Check operation of keeper by fully opening and closing.		
Snap fastener	Any fastener not functioning properly, i.e., fails to snap closed, provide a secure closure, or open freely.	108	
	NOTE: The fasteners shall be snapped and unsnapped twice to determine whether parts of fastener separate freely and also effect a secure closure.		
	Clinched excessively tight, cutting adjacent material.	109	
	Clinched loosely, permitting any component to rotate freely but not to the degree that any component can be expected to become detached during use.		204
	Clinched loosely to the degree that components can be expected to become detached during use.	110	
	NOTE: Incomplete roll of end of button or eyelet barrel is evidence of improper and insecure clinching.		
	Incorrect style.	111	
	More than three (3) splits in eyelet or button barrels.		205

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Table I: Classification Of Defects - Continued

Examine	Defect	Classification	
		Major	Minor
Seams and stitching: Open seams	Less than ½ inch.		206
	½ inch or more.	112	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two (2) or more consecutive skipped or runoff stitches occur.		
Raw edges (except where required)	More than ½ inch when securely caught in stitching.		207
	NOTE: Raw edges not securely caught in stitching shall be classified as open seams		
Seams and stitch types	Wrong seam or stitch type.	113	
Stitch tension	Loose, resulting in loose top or bobbin thread.		208
	Excessively tight, resulting in puckering of material.		209
Stitches per inch	One (1) stitch less than minimum specified. 1/		210
	Two (2) or more stitches less than minimum specified. 1/	114	
	One (1) or more stitches in excess of maximum specified. 1/		211
Stitching ends (on stitch type 301)	Not secured as specified (except where ends are held down by other stitching or turned under in a hem).		212
Thread breaks, skipped stitches or runoffs (on type 301 stitching)	Thread breaks, or two (2) or more consecutive skipped or runoff stitches overstitched less than ½ inch in each direction beyond the defective stitching area		213
	NOTE: Thread breaks, or two (2) or more consecutive skipped or runoff stitches not overstitched shall be classified as open seams.		
Bartacks	Loose stitching, incomplete or broken.		214
	Any omitted.	115	
Rows of stitching	Any row missing except on keeper retainer.	116	
	On keeper retainer:		
	-One (1) row of stitching omitted.		215
	-Two (2) or more rows of stitching omitted.	117	
Cleanness	Grease or oil stains clearly noticeable.		216
	Thread ends not trimmed throughout as specified.		217

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Table I: Classification Of Defects - Continued

Examine	Defect	Classification	
		Major	Minor
Components and assembly	Any component or required operation omitted (unless otherwise classified herein).	118	
	Mend, darn or patch.	119	
	Needle chews:		
	- up to 1/8 inch in length.		218
	- 1/8 inch or more in length.	120	
Binding tape	Loosely applied but not exposing raw edge of material.		219
	Loosely applied exposing raw edge of material.	121	
Fastener tape	Location of hook and pile reversed, i.e., hook attached where pile is required or vice versa.		220
Identification marking	Omitted, incorrect, illegible, misplaced, or size of characters not as specified.		221

1/ Defect to be scored when condition exists on major portion of seam. Applicable to each individual seam.

4.4.3.2 Dimensional examination. Examination shall be made for compliance with all dimensions shown on Drawing 2-2-282 which can be examined on the end item, including stitch margins and gage, excluding reference dimensions. Any dimension exceeding the applicable tolerance shall constitute a defect.

4.4.3.3 Visual shade matching (all Classes). The color and appearance of the case shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.6) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the case shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.6.1). The case shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The case is designed for carrying maintenance equipment for the M16A1 Rifle.

6.2 Acquisition requirements. Acquisition documents should specify the following:
a. Title, number and date of this specification.
b. When a first article sample is required (see 3.2, 4.3 and 6.4).

6.3 Samples. For access to samples, address the procuring activity issuing the invitation for bid.

6.4 First article. When a first article sample is required, it will be inspected and approved under the appropriate provisions of FAR 7-104.55. The first article should be a preproduction sample consisting of one completed case. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article.

6.5 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens".

6.5.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75.

6.6 Changes from previous issue. Due to the extent and the number of revisions made, no indications of changes (additions, modifications, corrections, deletions) from the previous issue were made. Bidders and contractors are cautioned to evaluate all of the requirements of this document based on the entire content of the document, due to the number and extent of revisions made to the previous issue.

Custodians:
Army – GL
Navy – MC

Preparing Activity:
DLA – CT

Project Number: 8465-2024-001

Review Activities:
Army – MD

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